

1. Write a Java application to simulate a **Bank Account system**. The program should allow withdrawal from an account. If the withdrawal amount is greater than the available balance, the program should throw a **user-defined exception called `InsufficientBalanceException`** and display an appropriate message. Otherwise, complete the withdrawal and display the remaining balance. (Also demonstrate same using Arithmetic Exception)
2. Write a Java program to simulate an **Online Movie Ticket Booking System**. A theater has a limited number of tickets available for a movie show. Multiple customers attempt to book tickets at the same time.

Create a class `TicketBooking` that contains the total number of available tickets and a method `bookTicket(int num)` to book the required number of tickets. Since multiple customers (threads) may try to access the booking method simultaneously, use the **synchronized keyword** to ensure that only one thread can access the booking method at a time.

Create a `Customer` thread class that attempts to book tickets using the shared `TicketBooking` object. The program should display whether the booking was successful and the number of tickets remaining after each transaction.

Demonstrate the concept of **synchronization in multithreading** by creating at least two customer threads attempting to book tickets concurrently.